Summary Notes:

Monitoring and Logging in Kubernetes

Introduction

Overview

Here is an overview of the topics covered in the session:

1. **Introduction to Kubernetes Objects**
    - Overview of Kubernetes objects such as Pods, ReplicaSets, Deployments, Services, and Namespaces.

2. **ReplicaSets**
    - Purpose and configuration of ReplicaSets to maintain pod availability.

3. **Deployments**
    - Declarative updates and control over pods and ReplicaSets.

4. **Kubernetes Services and Types**
    - Overview of services in Kubernetes and different service types (ClusterIP, NodePort, LoadBalancer, ExternalName).

5. **Namespaces in Kubernetes**
    - Use of namespaces for virtual clustering and resource isolation.

6. **Managed Kubernetes Services**
    - Introduction to AWS EKS and its advantages.

7. **Node Management in Kubernetes**
    - Differences between self-managed nodes and managed node groups.

8. **AWS Fargate**
    - On-demand compute for Kubernetes containers.

© Copyright upGrad Education Pvt. Ltd. All rights reserved

9. **Cluster Authentication**
    - IAM roles and authentication in Kubernetes clusters.

10. **Autoscaling in Kubernetes**
    - Node autoscaling, Horizontal and Vertical Pod Autoscalers.

11. **Metric Server in Kubernetes**
    - Installation and usage for resource monitoring.

12. **IAM Roles and Policies**
    - Granting permissions using IAM roles for EKS resources.

Kubernetes Objects

- Kubernetes objects define the state and configuration of applications running in a Kubernetes cluster.
- **Pods:** The smallest deployable unit, representing an instance of a process.
- **ReplicaSets:** Ensure a specified number of pod replicas are running.
- **Deployments:** Declarative updates for pods and ReplicaSets.
- **Services:** Manage the access and networking for pods.
- **Namespaces:** Logical isolation of resources within the same Kubernetes cluster.

ReplicaSets

- Purpose: Maintain a stable set of pod replicas.
- Configuration:
    - Selector: Defines which pods the ReplicaSet manages.
    - Replicas: Specifies the number of pod instances to maintain.
    - Pod Template: The blueprint for creating new pods when replicas are required.

Deployments

- **Purpose:** Manage the lifecycle of ReplicaSets and ensure the desired state of the application.
- **Key Feature:** Allows scaling, updating, and rolling back pods smoothly without downtime.

Kubernetes Services and Types

- Service: Provides a way to expose a set of pods as a network service.
- Types of Services:
    - **ClusterIP:** Accessible only within the cluster.
    - **NodePort:** Exposes services on static ports on each node's IP.
    - **LoadBalancer:** Uses a cloud provider's load balancer to expose services externally.

© Copyright upGrad Education Pvt. Ltd. All rights reserved

- **ExternalName:** Maps a service to an external DNS name.

Namespaces in Kubernetes

- **Purpose:** Virtual clusters that allow multiple teams to share the same physical cluster while keeping resource names unique within their own namespaces.
- **Use Case:** Useful for organizing and managing resources in a multi-tenant environment.

Managed Kubernetes Services (AWS EKS)

- **Introduction to AWS EKS:** A managed Kubernetes service that eliminates the need to manage the Kubernetes control plane, scaling it automatically across AWS availability zones.
- Advantages:
  - **High Availability:** Provides a 99.95% SLA for uptime.
  - **Seamless Integration:** Integrated with AWS services like IAM, EC2, Fargate, and load balancers for easier scaling and security.
  - **Cost Efficiency:** Supports on-demand and serverless compute options.

Node Management in Kubernetes

- **Self-Managed Nodes:** You manually provision and register EC2 instances or autoscaling groups.
- **Managed Node Groups:** EKS automatically provisions EC2 instances as part of autoscaling groups and manages them on your behalf.

AWS Fargate

- **Overview:** Provides on-demand compute capacity for Kubernetes pods without needing to provision virtual machines.
- **Limitations:** Fargate does not support DaemonSets, and Fargate profiles must be defined.

Cluster Authentication

- **IAM Roles:** Control access to EKS clusters by using the aws-auth ConfigMap.
- **Permissions:** The user or role that creates the cluster has system
- permissions, and additional users or roles can be added to the aws-auth ConfigMap for access.

Autoscaling in Kubernetes

- **Node Autoscaling:** Kubernetes Cluster Autoscaler automatically adjusts the number of nodes in the cluster based on pod requirements.
- Pod Autoscaling:

© Copyright upGrad Education Pvt. Ltd. All rights reserved

- **Horizontal Pod Autoscaler (HPA):** Scales the number of pods based on CPU or memory utilization.
- **Vertical Pod Autoscaler (VPA):** Adjusts the CPU and memory limits for pods to optimize resource utilization.

Metric Server in Kubernetes

- **Purpose:** Collects resource usage data (e.g., CPU and memory) across the cluster to support autoscaling.
- Installation: Use the following script to install.

```
Unset
kubectl apply -f
https://github.com/kubernetes-sigs/metrics-server/releases/download/v0.3.6/comp
onents.yaml
```

IAM Roles and Policies

- **IAM Role Setup:** Required to manage EKS resources. Attach roles to EC2 instances or autoscaling groups to grant permissions.
- **Configuration:** Managed through the aws-auth ConfigMap in Kubernetes.

© Copyright upGrad Education Pvt. Ltd. All rights reserved